

L21: The (s,S) Inventory Control Policy

Simulation Without a Black Box

Outline

Learning Objectives

Inventory System and Costs

(s,S) Policy Definition

Sawtooth Diagram

SimPy Container Implementation

Manual Trace Table

Performance Measures

Service Level vs. Inventory Cost

Key Takeaways

Learning Objectives

By the end of this lecture you will be able to:

- Identify the four cost components of an inventory system.
- State the (s, S) policy and explain why it is optimal for the periodic-review model.
- Trace the on-hand inventory level through the first five periods manually.
- Implement the (s, S) policy in SimPy using a Container.
- Compute fill rate, mean on-hand, and total cost from simulation output.

Inventory System: Four Cost Components

Cost structure

Holding cost h Cost per unit per period of carrying inventory (capital, storage, spoilage).

Ordering cost K Fixed cost per order placed (admin, shipping, setup).

Unit cost c Variable cost per unit ordered.

Shortage cost p Cost per unit of unmet demand (lost sales, backlog penalty).

Total expected cost per period

$$C = K \cdot \mathbb{E}[\text{order}] + c \cdot q + h \cdot \bar{I}^+ + p \cdot \bar{I}^-$$

where $\bar{I}^+ = E[\max(I, 0)]$ (avg on-hand)
and $\bar{I}^- = E[\max(-I, 0)]$ (avg backlog).

Simulation estimates $\bar{I}^+$ and $\bar{I}^-$ as
time averages or period averages.

The (s,S) Inventory Policy

(s,S) policy (“min-max” or “two-bin”)

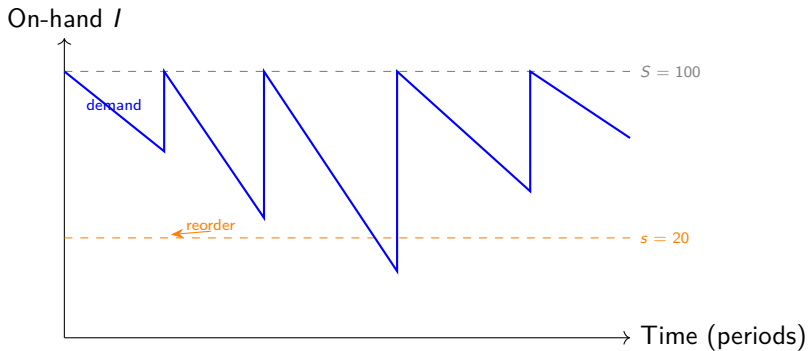
At each review epoch:

- If on-hand inventory $I \leq s$ (reorder point): **order** $q = S - I$ units (up to S).
- If $I > s$: **do not order**.

The order arrives after lead time L .

- **Special cases:**
 - $(s, S) = (s, s + Q)$: fixed order quantity Q (economic order quantity model).
 - $(s, S) = (-1, S)$: order-up-to (base-stock) policy.
- **Optimality:** (s, S) is optimal for periodic-review stochastic demand with fixed ordering cost $K > 0$ and continuous holding/shortage costs.
- Choosing s and S : balance holding cost (large S) vs. ordering frequency (large $s - s$ gap) vs. service level.

Sawtooth Pattern: On-Hand Inventory Over Time



Each sawtooth: inventory depletes under random demand until it hits s ; an order of size $S - I$ is placed; after lead time L , inventory jumps back toward S .

SimPy Container: (s,S) Policy Implementation

```
import simpy, numpy as np

s, S, L = 20, 100, 2      # reorder pt, order-up-to, lead time
h, K, shortage_cost = 0.5, 50, 5.0

def demand_process(env, inv, rng, stats):
    while True:
        yield env.timeout(1.0)          # daily review
        d = rng.poisson(lam=8)          # random daily demand
        fill = min(d, inv.level)
        yield inv.get(fill)
        stats["demand"] += d
        stats["filled"] += fill
        stats["short"] += d - fill
        stats["holding"] += inv.level * h

def reviewer(env, inv, rng):
    pending = False
    while True:
        yield env.timeout(1.0)
        if inv.level <= s and not pending:
            pending = True
            order_qty = S - inv.level
            yield env.timeout(L)          # lead time
            yield inv.put(order_qty)
            pending = False
```

Manual Trace: First 5 Periods

Parameters: $s = 20$, $S = 100$, $L = 1$ day, starting $I_0 = 100$.

Day	Begin I	Demand	End I	Order?	Order qty
1	100	12	88	No	—
2	88	15	73	No	—
3	73	18	55	No	—
4	55	40	15	Yes	$S - 15 = 85$
5	15	8	7	Yes	$S - 7 = 93$
(+1)	$7 + 85 = 92$	11	81	No	—

Day 4: inventory hits $15 \leq s = 20$; order 85 units, arrives day 5 (lead time 1). Day 5 starts with replenishment already counted in “Begin I ” column above.

Performance Measures from Simulation Output

Measure	Computation
Fill rate (CSL)	$\text{filled} / \text{demand}$
Avg on-hand $\bar{I}$	$\text{holding} / (h * T)$
Stockout rate	$\text{short} / \text{demand}$
Orders placed	count of replenishment events
Total cost	$K \cdot n_{\text{orders}} + h\bar{I}T + p \cdot \text{short}$

Cycle-service level (CSL)

$$\text{CSL} = P[\text{no stockout in a cycle}]$$

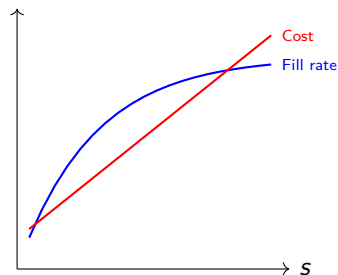
Vs. **fill rate**:

$\text{FR} = E[\text{demand met}] / E[\text{demand}]$. Fill rate is usually higher than CSL — they answer different questions.

Trade-off: Service Level vs. Inventory Cost

Increasing s (higher reorder point) raises fill rate but also raises average on-hand inventory (holding cost).

s	Fill rate	$\bar{I}$	Total cost
10	91%	42	312
20	96%	51	345
30	99%	60	378
40	99.7%	70	415



Choose s at the “knee” where marginal cost exceeds marginal service gain.

Key Takeaways

Core ideas from L21

1. The (s, S) policy is: order up to S whenever on-hand $\leq s$.
2. Four costs drive inventory decisions: holding, ordering (fixed), unit, and shortage.
3. SimPy **Container** models inventory level; `get` = demand, `put` = replenishment.
4. The sawtooth diagram visualises the inventory cycle and identifies stockout risk.
5. **Fill rate** and **CSL** answer different questions — report both.

Next: L22 — Comparing (s,S) policies: scenario analysis, CRN, and confidence intervals.